

ECE 315: Signals and Systems

Homework 9

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This homework is due on **5/4, 11:59pm**. For analytical problems, write your solutions independently by hand. For coding problems, attach your source code and answers. Combine them in one PDF document and submit on CANVAS. Please ensure that your PDF is clearly readable on a computer.

1. (2pts: Fourier transform) Consider the following discrete-time signals:

- $x[n] = \frac{1}{2^{|n|}} u[-n - 1]$;
- $x[n] = \sin(\frac{\pi}{2}n) + \cos(n)$;
- $x[n] = (n - 1) \frac{1}{2^{|n|}}$;
- $x[n] = \begin{cases} n & -2 \leq n \leq 2 \\ 0 & \text{otherwise.} \end{cases}$

Compute the Fourier transform $X(e^{jw})$ for each discrete-time signal.

2. (2pts: Inverse Fourier transform) Consider the following Fourier transforms:

- $X(e^{jw}) = e^{-jw/2}$ for $|w| \leq \pi$;
- $X(e^{jw}) = \cos^2(w)$;
- $X(e^{jw}) = \sum_{k=-\infty}^{\infty} (-1)^k \delta(w - \frac{\pi}{2}k)$;
- $X(e^{jw}) = \frac{e^{-jw} - \frac{1}{5}}{1 - \frac{1}{5}e^{-jw}}$

Compute the discrete-time signal $x[n]$ for each Fourier transform.

3. (2pts: Fourier transform properties) Consider the following two discrete-time signals:

- (i) $x[n] = \frac{1}{2^{|n|}}$
- (ii) $x[n] = \delta[n - 1] + \delta[n + 2]$.

Without explicitly computing $X(e^{jw})$, apply the Fourier transform properties to determine whether the Fourier transform has the following quantities:

- (i) $X(e^{jw})$ periodic;
- (ii) $X(e^{j0}) = 0$;
- (iii) $\int_{-\pi}^{\pi} X(e^{jw}) dw = 0$;
- (iv) $X(e^{jw})$ is purely real;
- (iv) $X(e^{jw})$ is purely imaginary.

Justify your choices.

4. (1pt: Fourier transform, convolution) Consider a system consisting of the cascade of two discrete-time linear-time invariant systems with frequency responses

$$H_1(e^{jw}) = \frac{2 - e^{-jw}}{1 + \frac{1}{2}e^{-jw}}$$

$$H_2(e^{jw}) = \frac{1}{1 - \frac{1}{2}e^{-jw} + \frac{1}{4}e^{-j2w}}$$



- (i) Find the difference equation describing the input and output $x[n]$ and $y[n]$;
 - (iii) Determine the impulse response of the overall system.
5. (1.5pts: LTI systems by difference equations) Consider the following discrete-time causal linear-time invariant system with input $x[n]$ and output $y[n]$ that is described by a difference equation:

$$y[n] + \frac{1}{2}y[n-1] = x[n].$$

- (i) Determine the frequency response $H(e^{jw})$ of this system;
 - (ii) Compute the outputs of the LTI system for two inputs: $x[n] = \frac{1}{2^n}u[n]$ and $x[n] = \delta[n] - \frac{1}{2}\delta[n-1]$;
 - (ii) Compute the outputs of the LTI system for two inputs with Fourier transforms: $X(e^{jw}) = \frac{1 - \frac{1}{4}e^{-jw}}{1 + \frac{1}{2}e^{-jw}}$ and $X(e^{jw}) = 1 + 2e^{-3jw}$.
6. (1.5pts: LTI systems by difference equations) Consider the following discrete-time linear-time invariant system with input $x[n]$ and output $y[n]$:

$$x[n] = \frac{1}{2^n}u[n] - \frac{1}{4} \frac{1}{2^{n-1}}u[n-1] \text{ and } y[n] = \frac{1}{3^n}u[n]$$

- (i) Determine the impulse response $h[n]$ of this system;
- (ii) Determine the frequency response $H(e^{jw})$ of this system;
- (iii) Find the difference equation relating the input and the output of this system.

7. Read Section **9.7-1**, and complete the problems **9.1-1** and **9.1-2** in **LSS**.